

PROFESSIONAL EXPERIENCE

Head of Research <i>Tutor Intelligence</i>	Jun 2026 — Present <i>Watertown, MA</i>
- Lead robot learning research linking policy training, demonstration quality, and deployment on a 100-robot bimanual fleet - Led team improvements to full-rate, synchronized data logging and headless teleoperation; co-authored RSS 2026 workshop paper on demonstration quality - Moved research training to a 32-H100 allocation and self-hosted MLflow; designed Rust middleware for data collection and policy execution	
Research Lead	Sep 2023 — Jun 2026
Research Scientist <i>RAI Institute (formerly Boston Dynamics AI Institute)</i>	Dec 2022 — Aug 2023 <i>Cambridge, MA</i>
- Built Foundation Models (10+ researchers) and Capture (30 people) teams, and 320-H100 training infrastructure - Led co-design of handheld and robotic grippers for force-based data collection and learned manipulation (Koala, 2026) - Developed video prediction and multimodal models, and shared policy execution infrastructure used across research teams - Established research partnerships with Google, Columbia, ETH Zurich, and Agile Robots	
Research Fellow <i>Army Research Lab</i>	Sep 2018 — Jul 2022 <i>Aberdeen, MD</i>
- Won 1st place MineRL BASALT Competition (NeurIPS 2021): combined imitation learning, human preference data, and hierarchical knowledge engineering to solve tasks defined only by natural language - Developed mobile manipulation approaches without runtime localization (IROS 2022 Best Paper Finalist) - Developed multimodal geometric learning fusing vision and tactile sensing for robotic grasping (ICRA 2019)	

SELECTED RESEARCH

BRIDGE: Contact-Rich Manipulation <i>RAI Institute, under review at ICRA 2027</i>	2026 <i>Cambridge, MA</i>
- Co-authored state-gated diffusion policy experts that combine handheld demonstrations with targeted teleoperation - Improved task success over handheld-only baselines across three contact-rich manipulation tasks	
Koala: Gripper and Data-Capture Co-Design <i>RAI Institute, under review at IEEE T-RO</i>	2026 <i>Cambridge, MA</i>
- Led co-design of handheld and actuated grippers with matched geometry, sensing, and force transmission - Integrated the platform with data collection and learned policy execution for dexterous manipulation	
DF1: Demonstration Quality at Fleet Scale <i>Tutor Intelligence, RSS 2026 Workshop: It's the Demos</i>	2026 <i>Watertown, MA</i>
- Co-authored a demonstration-quality pipeline for a 100-robot bimanual fleet - Connected teleoperation, data curation, and policy training with human interventions to correct rollout failures	

SELECTED PUBLICATIONS

1. Hajj-Ahmad et al., Watkins . "Koala Gripper: Co-designing Robotic Grippers and Data-Capture Devices for Scaling Dexterous Manipulation Learning." <i>IEEE Transactions on Robotics (T-RO)</i> , under review	
2. Surendran, Peri, Watkins . "Bridging Handheld and Teleoperated Supervision for Contact-Rich Manipulation via State-Gated Experts." <i>ICRA 2027</i> , under review	
3. Kapoor et al., Watkins . "Maintaining Demonstration Quality in a 100-Robot Teleoperation Pipeline." <i>RSS 2026 Workshop: It's the Demos</i>	
4. Shang et al., Watkins , Herlant. "Theia: Distilling Diverse Vision Foundation Models for Robot Learning." <i>CoRL 2024</i>	

EDUCATION

Ph.D. in Computer Science , <i>Columbia University</i>	2017 — 2022
Thesis: <i>Learning Mobile Manipulation</i> Advisor: Prof. Peter Allen Army Research Lab Fellow	
B.S./M.S. in Computer Science , <i>Columbia University</i> , 4.0 GPA (M.S.)	2012 — 2017

SKILLS

ML / Robotics	Foundation Models, Diffusion Policies, Imitation Learning, RLHF, Manipulation
Systems	Python, C++, Rust, PyTorch, CUDA, ROS 2, Distributed Training, GPU Infrastructure
Communication	English, Spanish